```
import pandas as pd
import numpy as np
from sklearn.preprocessing import MinMaxScaler
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense
import matplotlib.pyplot as plt

# 1. Load your dataset
data = pd.read_csv('LOYALTY.csv')

# 2. Select your 10 SEM-identified features
selected_features = ['ONR', 'C', 'SM', 'T', 'PU',
                     'SQ', 'CS', 'Csat', 'CL', 'PI']

X = data[selected_features].values
y = data['LOYALTY'].values

# 3. Scale the data
scaler_X = MinMaxScaler()
scaler_y = MinMaxScaler()

X_scaled = scaler_X.fit_transform(X)
y_scaled = scaler_y.fit_transform(y.reshape(-1, 1))

# 4. Create sequences for LSTM
def create_sequences(X, y, time_steps=10):
    Xs, ys = [], []
    for i in range(time_steps, len(X)):
        Xs.append(X[i-time_steps:i])
        ys.append(y[i])
    return np.array(Xs), np.array(ys)

time_steps = 10
X_seq, y_seq = create_sequences(X_scaled, y_scaled, time_steps)

# 5. Split into training and testing sets
split_idx = int(0.8 * len(X_seq))
X_train, X_test = X_seq[:split_idx], X_seq[split_idx:]
y_train, y_test = y_seq[:split_idx], y_seq[split_idx:]

# 6. Build the LSTM model
model = Sequential()
model.add(LSTM(50, return_sequences=True, input_shape=(X_train.shape[1], X_train.shape[2])))
model.add(LSTM(50))
model.add(Dense(1))

model.compile(optimizer='adam', loss='mse')

# 7. Train the model
history = model.fit(X_train, y_train, epochs=50, validation_split=0.1, batch_size=32, verbose=1)
from tensorflow.keras.callbacks import EarlyStopping

early_stop = EarlyStopping(monitor='val_loss', patience=5, restore_best_weights=True)

model.fit(X_train, y_train, epochs=50, validation_split=0.1, batch_size=32,
          callbacks=[early_stop], verbose=1)
# 8. Predict and evaluate
y_pred = model.predict(X_test)
y_pred_inv = scaler_y.inverse_transform(y_pred)
y_test_inv = scaler_y.inverse_transform(y_test.reshape(-1, 1))

mae = mean_absolute_error(y_test_inv, y_pred_inv)
rmse = np.sqrt(mean_squared_error(y_test_inv, y_pred_inv))
r2 = r2_score(y_test_inv, y_pred_inv)

print(f"MAE: {mae:.4f}")
print(f"RMSE: {rmse:.4f}")
print(f"R²: {r2:.4f}")

# 9. Plot predictions
plt.figure(figsize=(10, 5))
plt.plot(y_test_inv, label='Actual Loyalty')
plt.plot(y_pred_inv, label='Predicted Loyalty')
plt.title('Actual vs Predicted Loyalty')
plt.xlabel('Time')
```

```
plt.ylabel('Loyalty Score')
plt.legend()
plt.grid(True)
plt.show()
plt.savefig('actual_vs_predicted_loyalty.png', dpi=300, bbox_inches='tight')
plt.savefig('training_loss_plot', dpi=300, bbox_inches='tight')

# 10. Export results
metrics_df = pd.DataFrame({
    'Metric': ['MAE', 'RMSE', 'R-squared'],
    'Value': [mae, rmse, r2]
})
metrics_df.to_csv('model_evaluation_metrics.csv', index=False)

predictions_df = pd.DataFrame({
    'Actual': y_test_inv.flatten(),
    'Predicted': y_pred_inv.flatten()
})
predictions_df.to_csv('model_predictions.csv', index=False)

model.save('loyalty_lstm_model.h5')
```

```
/usr/local/lib/python3.11/dist-packages/keras/src/layers/rnn/rnn.py:200: UserWarning: Do not pass an `input_shape`/`input_dim` argument
  super().__init__(**kwargs)
Epoch 1/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 5s 109ms/step - loss: 0.0630 - val_loss: 0.0100
Epoch 2/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 28ms/step - loss: 0.0348 - val_loss: 0.0138
Epoch 3/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0274 - val_loss: 0.0151
Epoch 4/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0251 - val_loss: 0.0102
Epoch 5/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 26ms/step - loss: 0.0327 - val_loss: 0.0106
Epoch 6/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0225 - val_loss: 0.0119
Epoch 7/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0242 - val_loss: 0.0122
Epoch 8/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0236 - val_loss: 0.0115
Epoch 9/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step - loss: 0.0249 - val_loss: 0.0111
Epoch 10/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0246 - val_loss: 0.0115
Epoch 11/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0247 - val_loss: 0.0103
Epoch 12/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0249 - val_loss: 0.0109
Epoch 13/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0230 - val_loss: 0.0110
Epoch 14/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0266 - val_loss: 0.0112
Epoch 15/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0268 - val_loss: 0.0097
Epoch 16/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0218 - val_loss: 0.0117
Epoch 17/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0252 - val_loss: 0.0098
Epoch 18/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0239 - val_loss: 0.0107
Epoch 19/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0229 - val_loss: 0.0096
Epoch 20/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0234 - val_loss: 0.0103
Epoch 21/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0212 - val_loss: 0.0094
Epoch 22/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0231 - val_loss: 0.0101
Epoch 23/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0235 - val_loss: 0.0101
Epoch 24/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0260 - val_loss: 0.0096
Epoch 25/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 36ms/step - loss: 0.0196 - val_loss: 0.0113
Epoch 26/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.0214 - val_loss: 0.0093
Epoch 27/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 38ms/step - loss: 0.0242 - val_loss: 0.0119
Epoch 28/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 37ms/step - loss: 0.0235 - val_loss: 0.0091
Epoch 29/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 40ms/step - loss: 0.0196 - val_loss: 0.0107
Epoch 30/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 1s 42ms/step - loss: 0.0206 - val_loss: 0.0090
Epoch 31/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 1s 24ms/step - loss: 0.0265 - val_loss: 0.0102
Epoch 32/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 30ms/step - loss: 0.0227 - val_loss: 0.0105
Epoch 33/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0251 - val_loss: 0.0087
Epoch 34/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 22ms/step - loss: 0.0221 - val_loss: 0.0088
Epoch 35/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step - loss: 0.0231 - val_loss: 0.0092
Epoch 36/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0204 - val_loss: 0.0089
Epoch 37/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0200 - val_loss: 0.0088
Epoch 38/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0247 - val_loss: 0.0100
Epoch 39/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0240 - val_loss: 0.0087
Epoch 40/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 23ms/step - loss: 0.0231 - val_loss: 0.0086
Epoch 41/50
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 24ms/step - loss: 0.0229 - val_loss: 0.0087
```